# Uninformed Search

## EECS 492

## Introduction to Artificial Intelligence

# Today's Class

- Search
  - Preliminary definitions
  - A general search architecture
  - Breadth-first search
  - Uniform cost search
  - Depth-first search and variations

# EECS 492 Topics

Boundaries are fuzzy!

## A.I.

- Search
  - Uninformed
  - Informed
  - Local
  - Adversarial
- Constraint Satisfaction
- Logic
  - Propositional
  - First-Order
- Game theory

## Machine Learning

- Decision Trees
- Linear Regression
- k-Nearest Neighbor
- SVMs
- Markov Decision Processes
- MCTS
- Reinforcement Learning

## Neural Networks

- Feed forward (& deep)
- Convolutional
- Recurrent
- Transformers
- Generative

Topics subject to modest change

Lots of math across the entire course!

# "Why not study only LLMs / modern GenAI / deep learning / neural networks?"

- Small models on personal machines, ok, but GenAI at scale is expensive
  - Cost to build yourself: millions to billions
  - Use another company's resources?
  - Significant energy / climate costs

7 trillion worldwide by 2030

If another technique can do the job, it will often be much cheaper and will require much less data.

- Neural networks are "black boxes", but in some domains, explainability is paramount.

Many techniques exist where the reason for a conclusion is more clear.

- Neural networks are often improved when combined with other techniques.
- High quality research is also done in other areas of AI, and this trickles down to business & public sector applications.

The field is constantly evolving. What's next?

- Neural network training is ***search***!

## NeurIPS 2024: Just a "few" examples

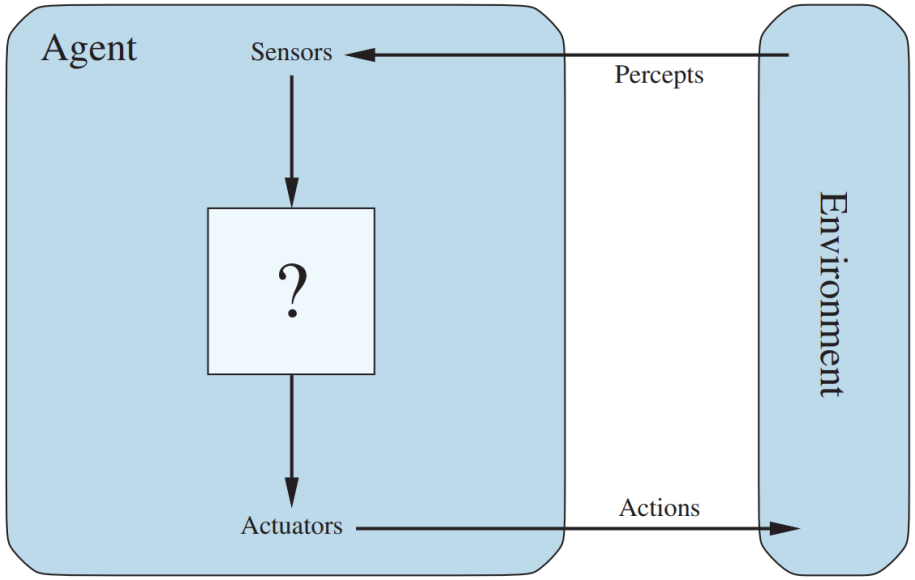
- 2024 proceedings
- All proceedings

- SeeA*: Efficient Exploration-Enhanced **A* Search** by Selective Sampling
- Double-Ended Synthesis Planning with Goal-Constrained **Bidirectional Search**
- Minimizing UCB: a Better **Local Search** Strategy in Local Bayesian Optimization
- Hybrid Top-Down Global Causal Discovery with **Local Search** for Linear and Nonlinear Additive Noise Models
- **Monte Carlo Tree Search** based Space Transfer for Black Box Optimization
- Generating Code World Models with Large Language Models Guided by **Monte Carlo Tree Search**
- ReST-**MCTS***: LLM Self-Training via Process Reward Guided Tree Search
- Speculative **Monte-Carlo Tree Search**
- Transition Constrained **Bayesian** Optimization via **Markov Decision Processes**

**And much more! See for yourself!**

# Back to Agents

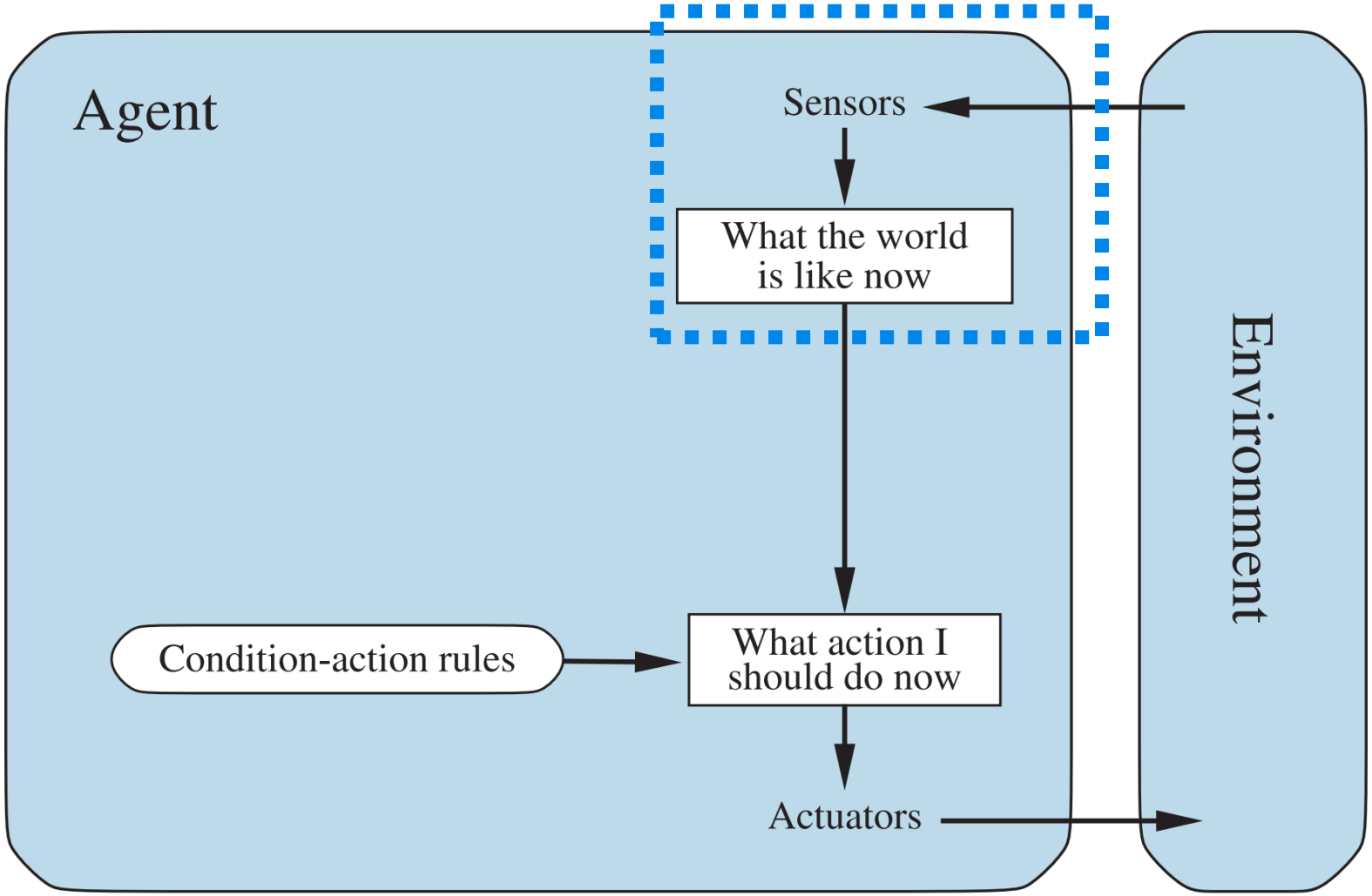
- An agent is anything that:
  - Perceives its environment through sensors
    - The things perceived are called *percepts*.
  - Decides how to act on its environment
    - Acts via *actuators*.




- An agent pursues a set of pre-specified goals, or a pre-specified strategy
  - Entrusted to act autonomously

# Simple Reflex Agent

React based on
**current percept only**.

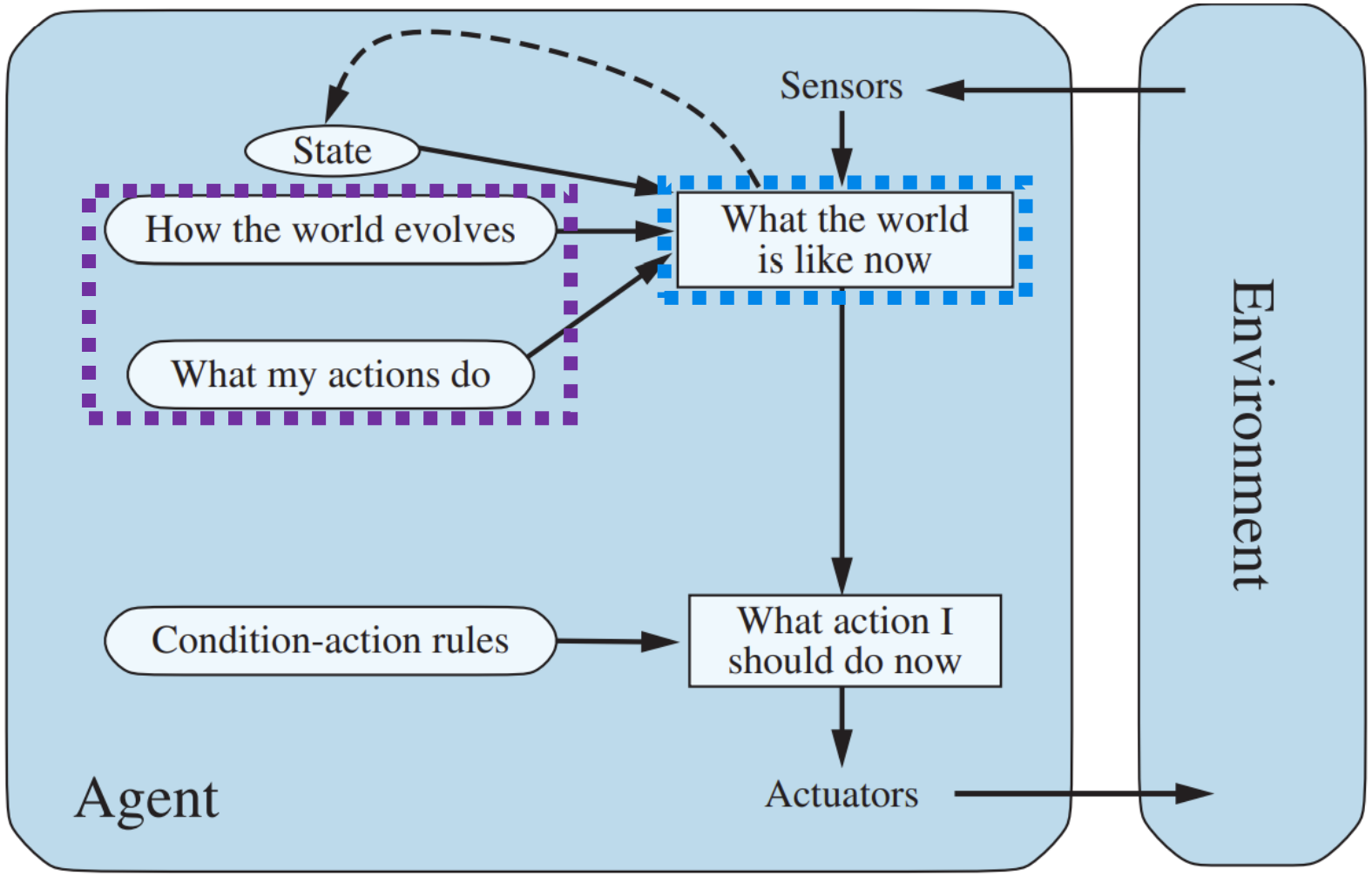


If floor is dirty below me, clean there.
else if …

# Model-Based Reflex Agent

Also:

- Info about **how the world changes**.
- Maintain **internal belief** of the state of the world right now.




I just cleaned in a spot, so the floor is no longer dirty there.

# Preliminary Definitions

# Problem Solving

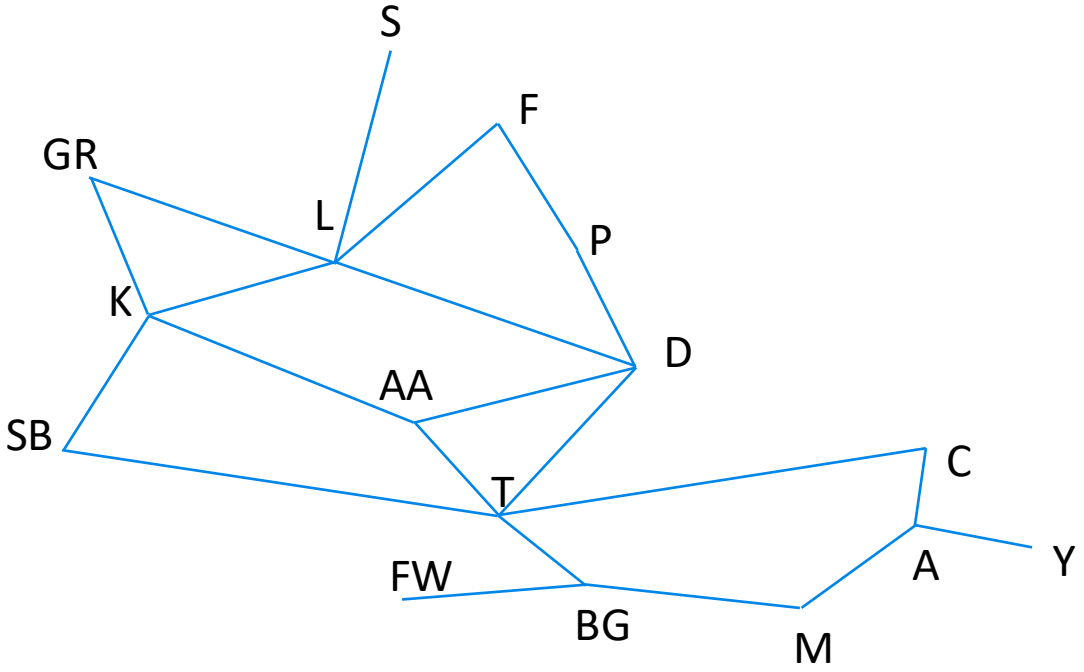
- An agent needs a way to choose the action(s) to take

- Now:
  - *Searching* for the best action(s) to take
    (for a model- and goal-based agent)

- Real-world problems
  - Route finding
  - VLSI layout (cell layout and channel routing)
  - Robotic manufacturing with partial ordering specified
  - Protein design (find sequence of amino acids that fold into a protein with desired properties)

- Let's start with an example, drawing general concepts from that

# Find a route from AA to SB




- Problem formulation:
  - State: The city I'm in
    - Example: in AA
  - Action: Drive to an adjacent city (map is known)
    - Example: AA → K
  - Only certain cities are considered
  - Only certain roads between cities are considered
- Initial state: in AA
- Goal state: in SB
- Search:
  - Agent *simulates* various actions, trying to find sequences to reach the goal.
  - Example:
    AA → T → SB
    AA → D →P → F → L → GR → K → SB
    …
  - Agent picks a sequence to then *execute*.

# Defining a search problem

1. **State space** –
   - Set of possible states

   Assume atomic.
2. **Initial state**
3. **Goal state(s)**
   - Single goal?
   - State with some characteristic?
     (ex: any city with average yearly temperature in the 70's)
4. **Actions**
   - `Actions(s)` is a finite set of actions possible from state `s`
5. **Transition model**
   - `Result(s, a)` returns the state resulting from taking action `a` in state `s`
6. **Action cost function**
   - `Action_Cost(s, a, s')` or `c(s, a, s')`

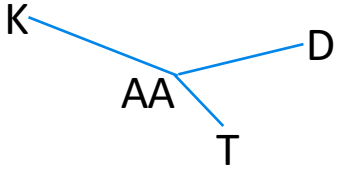
     Assume positive for now.

     Cost of doing action `a` when in state `s` to get to state `s'`.

In our example:

1. `{AA, SB, K, GR, L, S, F, P, D, T, FW, BG, M, A, C, Y}`
2. `AA`
3. `SB`

4. `Actions(AA) = {ToD, ToT, ToK}`




5. `Result(AA, ToK) = K`

   …

6. `Action_Cost(AA, ToK, K)` = …

   (miles, or time, or tolls, or how ugly the scenery is, or … any combination)

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     Cost of doing action `a` when in state `s` to get to state `s'`.

**Definitions**

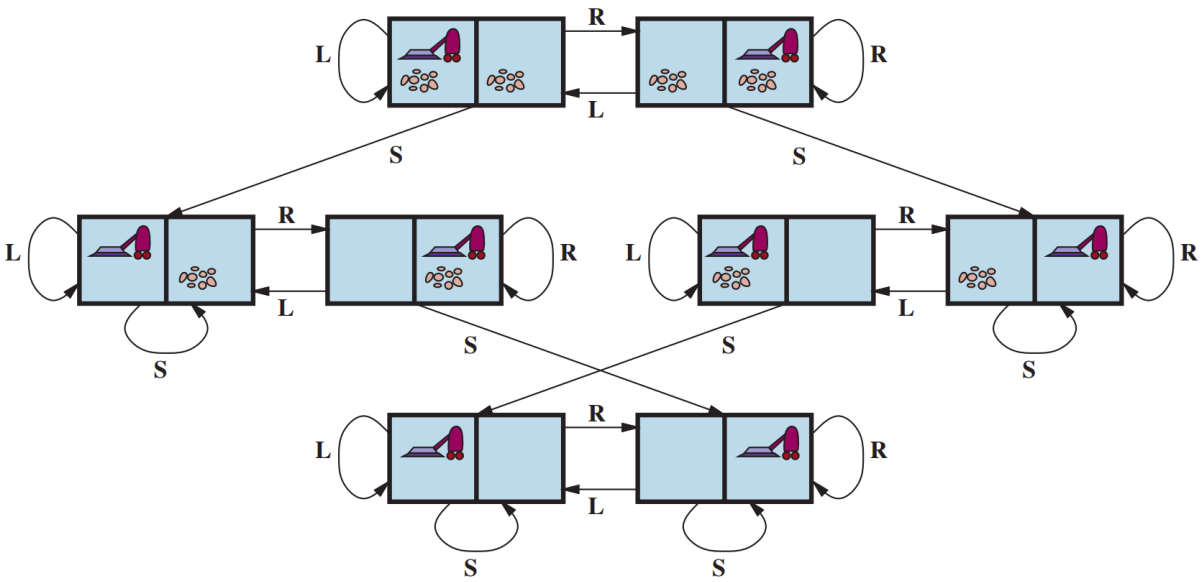
- **path** – start state, end state, sequence of actions (or, when no ambiguity, a sequence of states)
- **path cost** – sum of *action costs (step costs)* for each action in a path
- **solution** – a path from initial state to goal state
- **optimal solution** – solution with lowest path cost

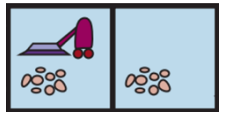
**Environment properties**:

- Fully observable, single agent, deterministic, sequential, static, discrete, known

# Defining a Search Problem: Vacuum

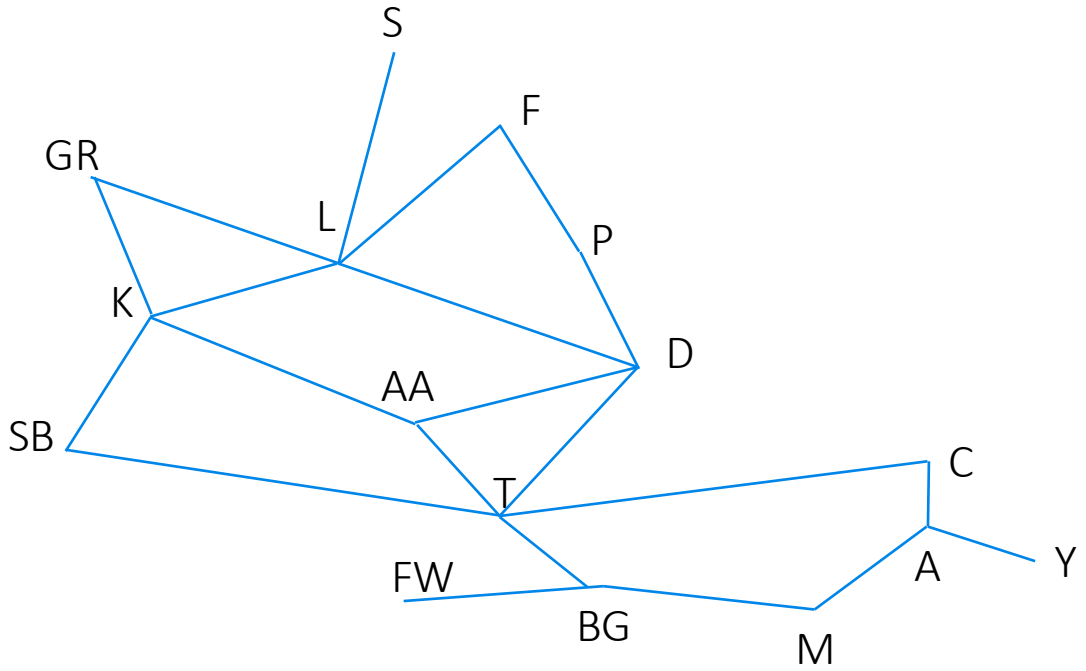
- Imagine just two possible locations:
- Each may or may not have dirt. Initially, they're both dirty.
- Vacuum can be at either spot. Initially, it's at the left spot.
- Vacuum can move or clean at current location.



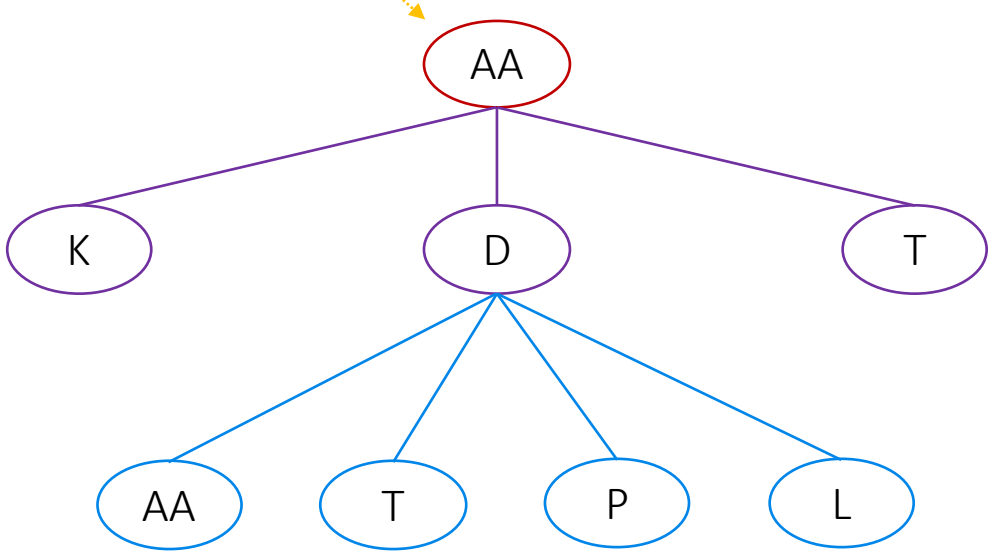

1. State space graph
2. Initial state – 
3. Goal state(s) – either state with both spots clean
4. Actions – L, R, S.
   (maybe disallow invalid moves)
5. Transition model – Defined by the state space graph.
6. Action cost function – Let's suppose all actions are equal cost here.

# Search Tree

The search algorithms we'll consider for now all operate by creating a search tree on a state-space graph.



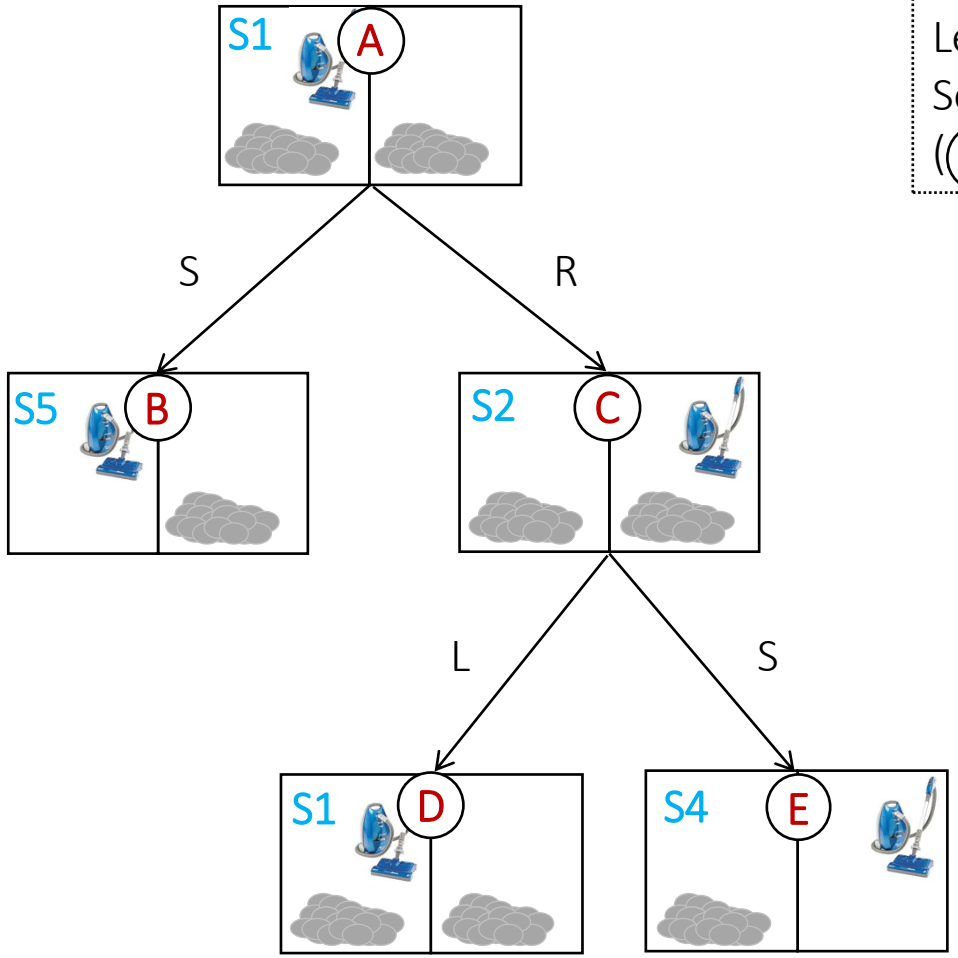

1) Initial state: AA



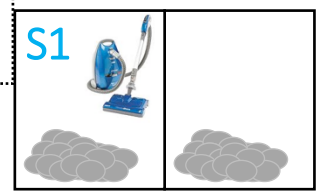

2) *Expand* AA (find all states reachable from AA with a single action)

3) Select (somehow…) another node at any level (K, AA, T, P, L, or T), and expand it…

# Vacuum: Search Tree




Let's label each node (A - E).
Some nodes contain the same state!
(A and D here both have state S1.)




Parent node also contains *its* parent node (the grandparent), and so on, to the root.

So each node has information about entire path from root (initial state).

To reduce notation, I won't always *show* this, though...

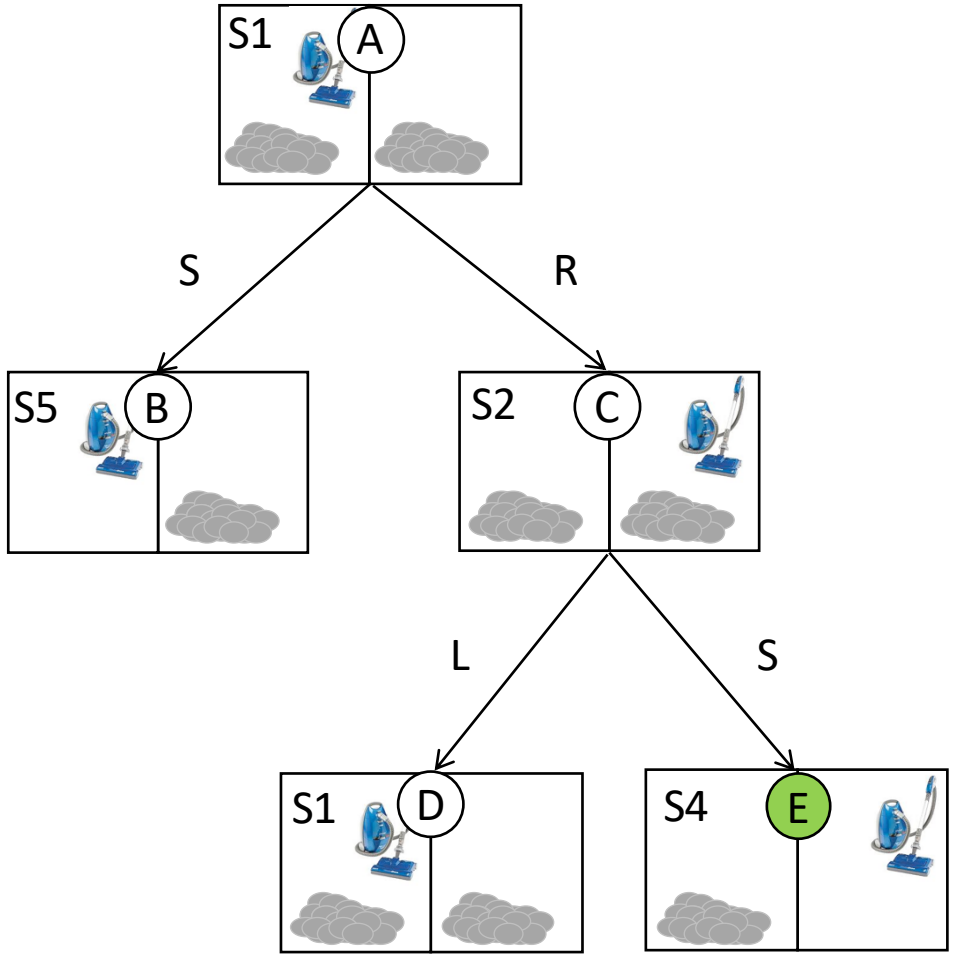
State – the way the world is
vs.
Node – part of the search tree

State and Node aren't the same thing!

- A Node contains:
  - State
  - Parent node
  - Action to get from parent to this node
  - path cost g(n) – cost from root node to this node

Omitting the non-move loops here, for brevity.

# Vacuum: Node Example




Let's assume each action (S, L, R) has the same cost. We'll just call that cost 1.

- Node E contains:
  - State **S4**
  - Parent **C**
  - Action **S**
  - g(n) = 2

- A Node contains:
  - State
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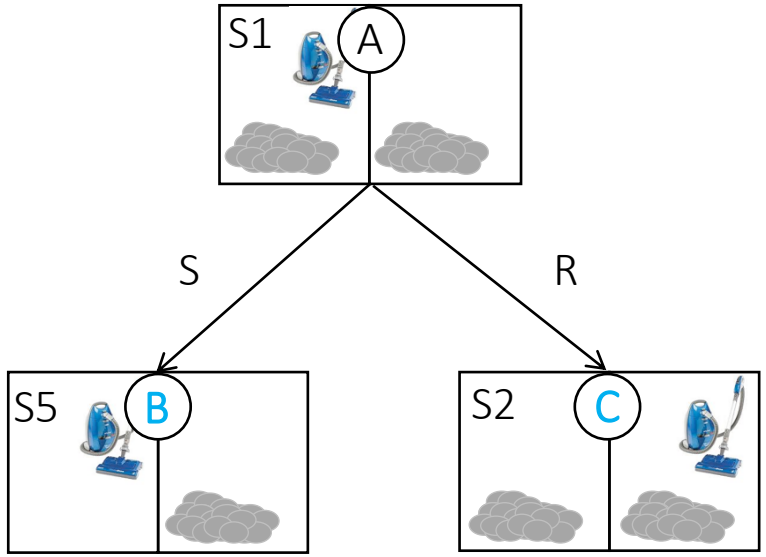
  - Action to get from parent to this node
  - path cost g(n) – cost from root node to this node

# Vacuum: Vocabulary




When node A was expanded, nodes B and C were generated.

- *expand* a node:
  - determine applicable actions
  - determine resulting states (*successors*)
  - *generate* nodes for each successor and add them to the search tree

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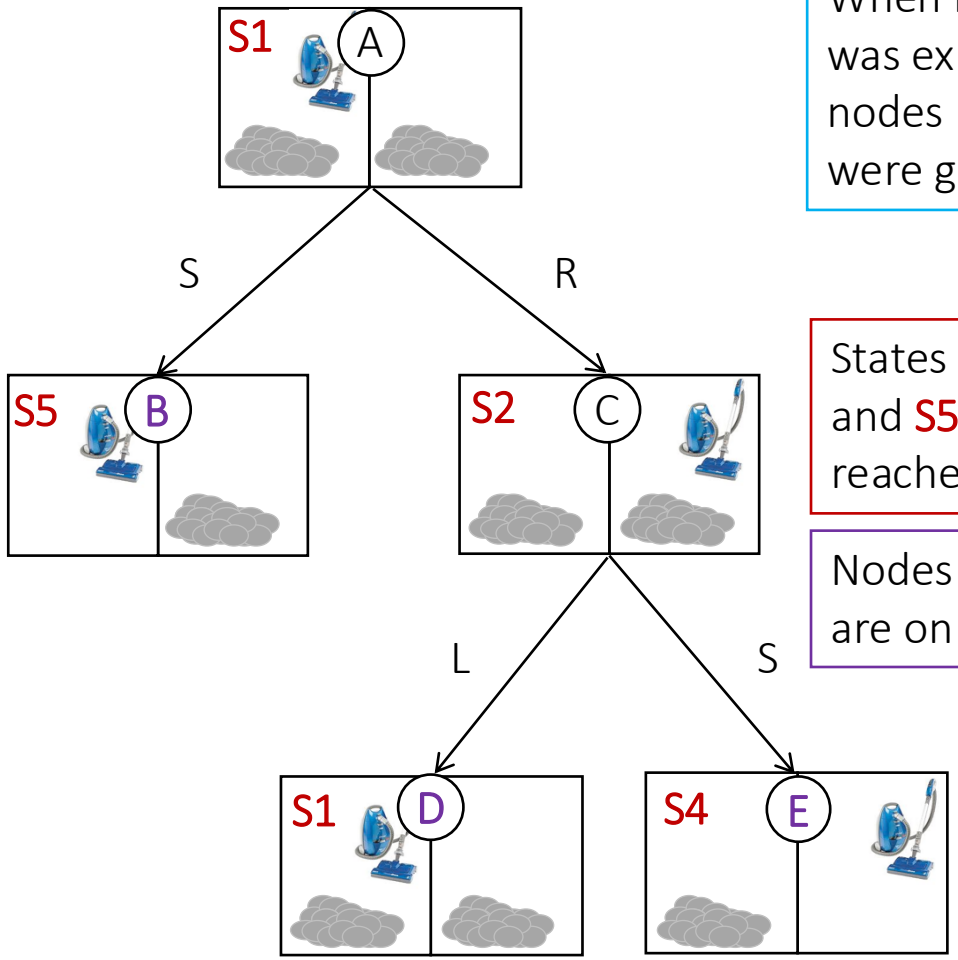



When node A was expanded, nodes B and C were generated.

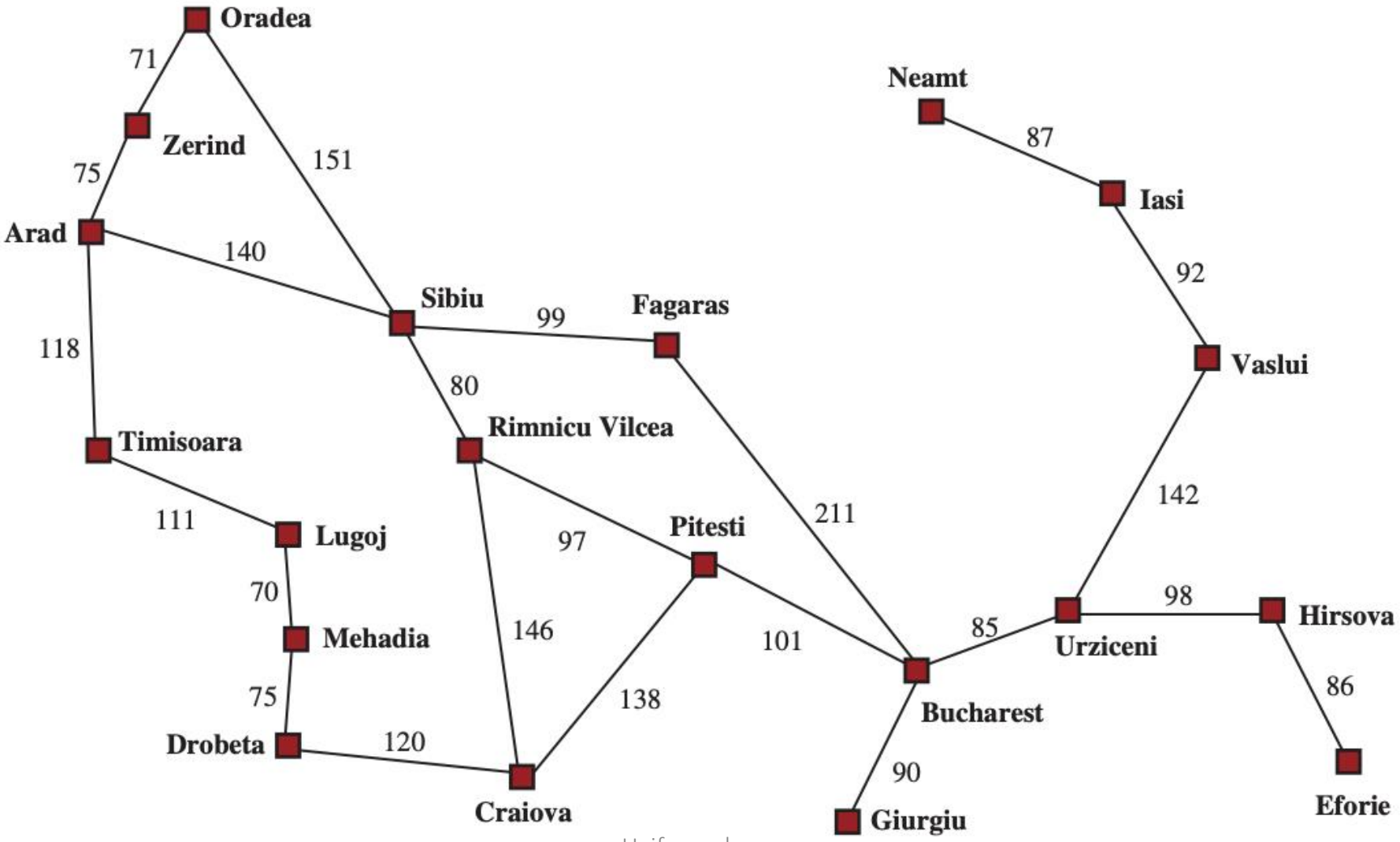
States S1, S2, S4, and S5 have been reached.

Nodes B, D, and E are on the frontier.

- *expand* a node:
  - determine applicable actions
  - determine resulting states (*successors*)
  - *generate* nodes for each successor and add them to the search tree
- A state has been ***reached*** if at least one node has been *generated* that contains it.
- A node is on the ***frontier*** if it has not been expanded yet. It's a leaf that hasn't been expanded (as opposed to a leaf that we did try to expand but that had no children).

- A Node contains:
  - State
  - Parent node
  - Action to get from parent to this node
  - path cost g(n) – cost from root node to this node

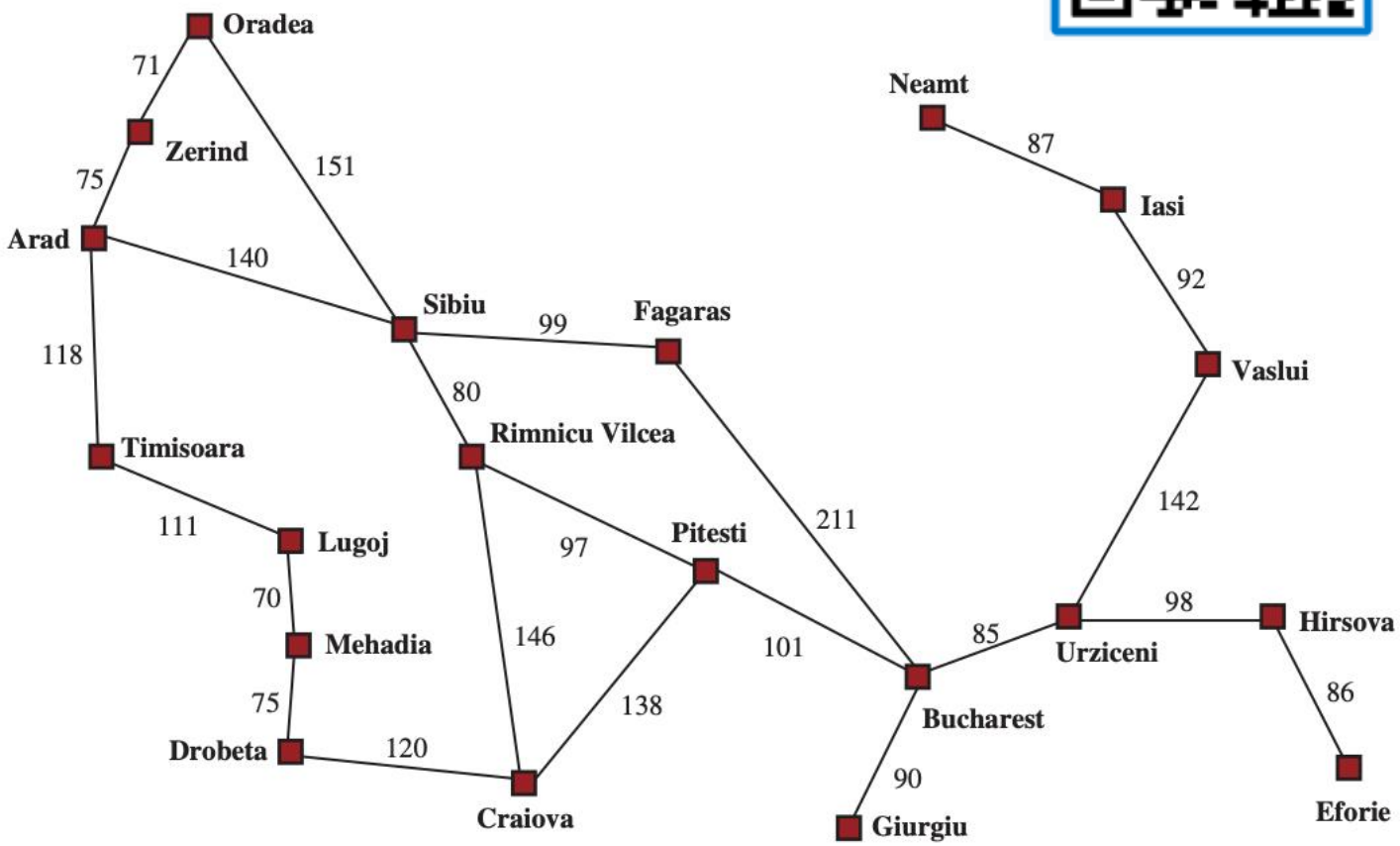
# Exercise: Toy Map of Romania (distances in miles)



# Exercise

When poll is active respond at [PollEv.com/492](PollEv.com/492)

- Let's say we're trying to get from Arad to Oradea
- What are the following quantities?
  - INITIAL-STATE
  - ACTIONS(Arad)
  - RESULT(Arad, Arad->Sibiu)
  - IS-GOAL(Sibiu)
  - ACTION-COST(Arad, Arad->Sibiu, Sibiu)



# Measuring Performance

- Completeness
  - Is the algorithm guaranteed to find a solution when there is one?
- Optimality
  - Does the strategy find the optimal solution?
  - Can define in terms of path cost
- Time complexity
  - How long does it take to find a solution?
- Space complexity
  - How much memory is needed to perform the search?

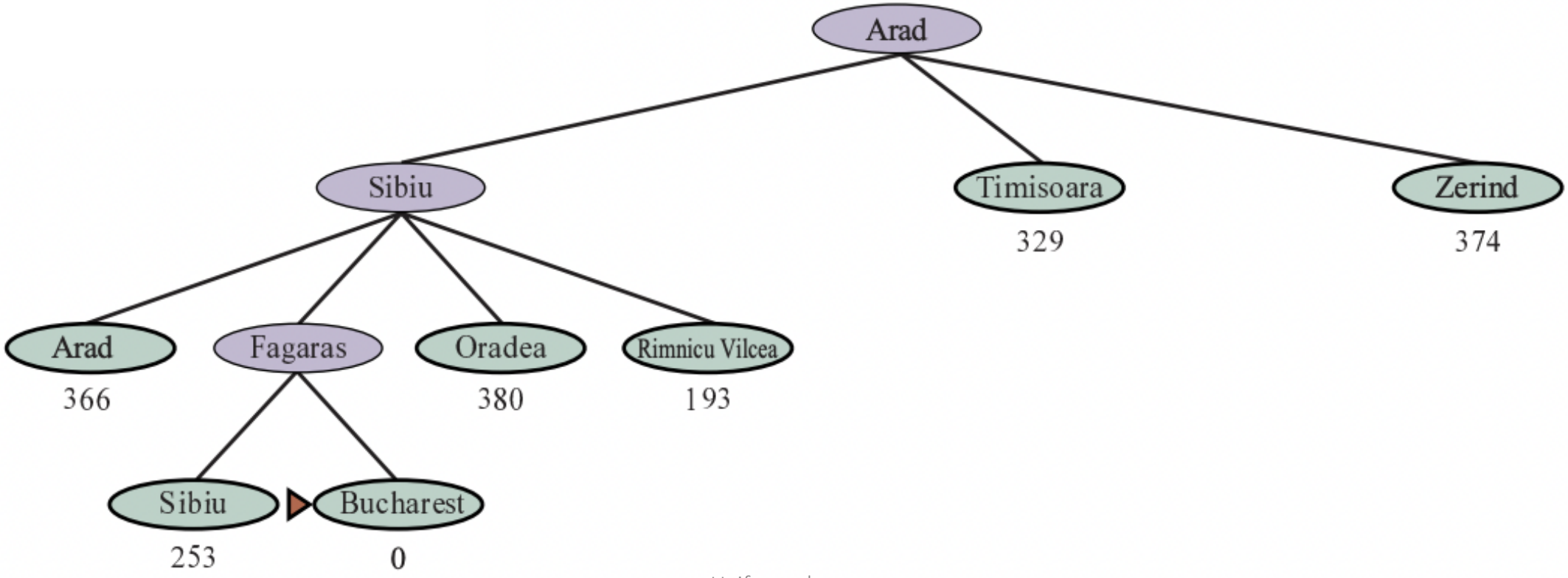
# Graph Properties

- Branching factor (b) = maximum number of successors for any node
- Depth (d) = shallowest goal node (number of actions in an optimal solution)
- Maximum path length (m) = longest length of any path in the search ~~space~~ tree (max number of actions in any path)

# Exercise

When poll is active respond at **PollEv.com/492**

Let's say we generate this search tree, searching from Arad to Bucharest. What is the branching factor (b), depth (d), and maximum path length (m) for this tree?

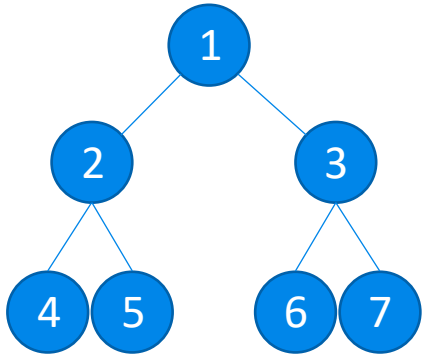


# A General Search Architecture

# Let’s think about a general search architecture

**function** BEST-FIRST-SEARCH(*problem*, f) **returns** a solution node or *failure*

The problem:




Frontier: []

Reached: []

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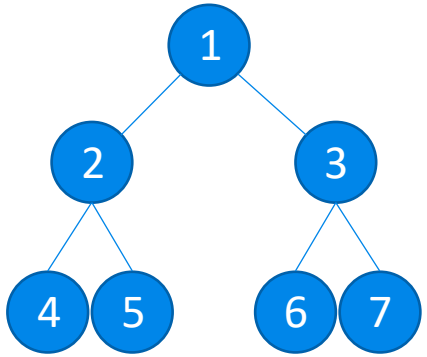


Frontier: []

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f is an evaluation function, we’ll talk about this at length

# Let’s think about a general search architecture

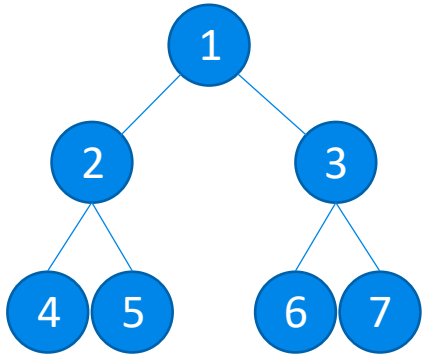
**function** BEST-FIRST-SEARCH(*problem*, f) **returns** a solution node or *failure*

*node* ← NODE(STATE=*problem*.INITIAL)

*frontier* ← a priority queue ordered by *f*, with *node* as an element

*reached* ← a lookup table, with one entry with key *problem*.INITIAL and value *node*

The problem:




Frontier: [1]

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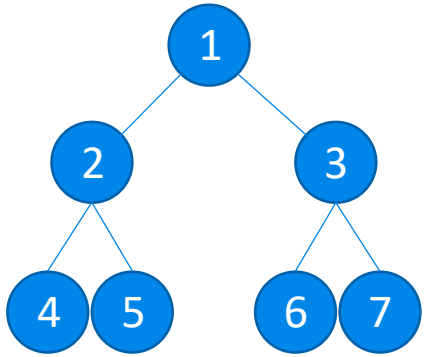
**while not** IS-EMPTY(*frontier*) **do**

The problem:

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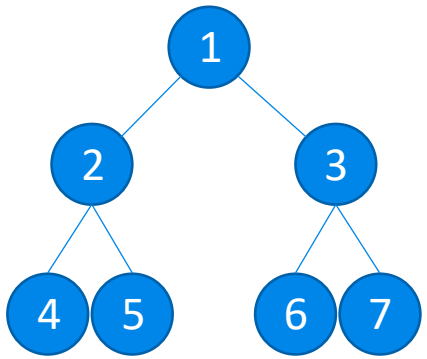


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We are now considering node 1

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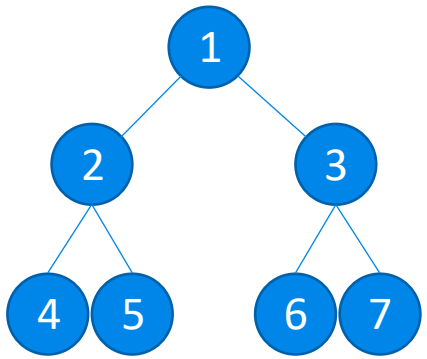


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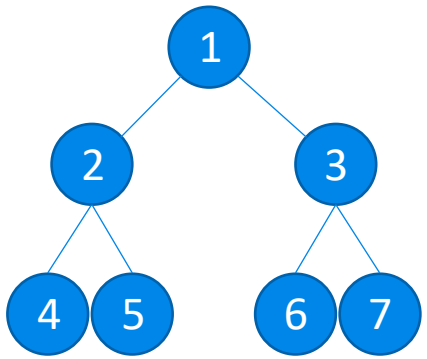
        if problem.IS-GOAL(node.STATE) then return node
```

The problem:




Frontier: []

Reached: [1]

Node 1 isn’t a goal node

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        for each child in EXPAND(problem, node) do
```
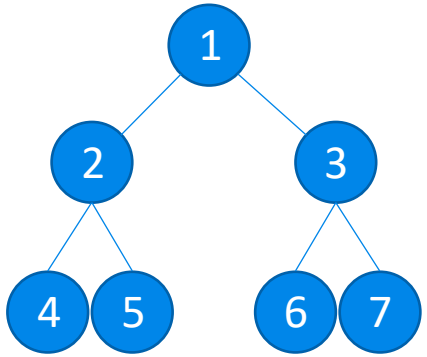

The problem:

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So, we'll expand it (find its children)

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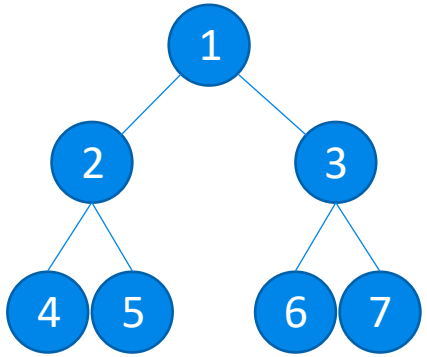
            s ← child.STATE
```

The problem:




Frontier: []

Reached: [1]

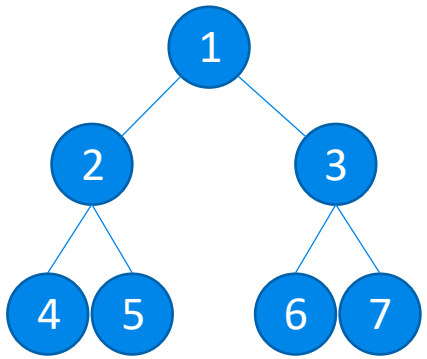
So, we'll expand it (find its children) – they are 2 and 3, we'll consider node 2

# Let's think about a general search architecture

```
function BEST-FIRST-SEARCH(problem, f) returns a solution node or failure
    node ← NODE(STATE=problem.INITIAL)
    frontier ← a priority queue ordered by f, with node as an element
    reached ← a lookup table, with one entry with key problem.INITIAL and value node

    while not IS-EMPTY(frontier) do
        node ← POP(frontier)
        if problem.IS-GOAL(node.STATE) then return node
        for each child in EXPAND(problem, node) do
            s ← child.STATE
```

The problem:




Frontier: []

Reached: [1]

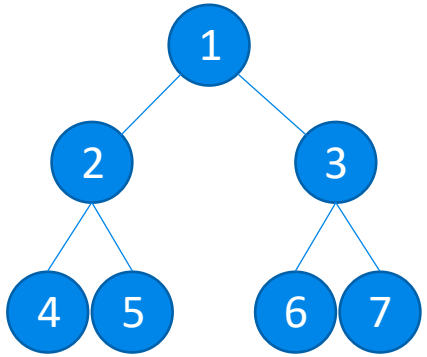
So, s is the state of node 2

# Let's think about a general search architecture

```
function BEST-FIRST-SEARCH(problem, f) returns a solution node or failure
    node ← NODE(STATE=problem.INITIAL)
    frontier ← a priority queue ordered by f, with node as an element
    reached ← a lookup table, with one entry with key problem.INITIAL and value node

    while not IS-EMPTY(frontier) do
        node ← POP(frontier)
        if problem.IS-GOAL(node.STATE) then return node
        for each child in EXPAND(problem, node) do
            s ← child.STATE
            if s is not in reached or child.PATH-COST < reached[s].PATH-COST then
                reached[s] ← child
                add child to frontier
```

The problem:




Frontier: [2]

Reached: [1, 2]

- If state s **not in** frontier, **add** child to frontier
- Else, **update** the node currently in frontier with the child node

# Let's think about a general search architecture

```
function BEST-FIRST-SEARCH(problem, f) returns a solution node or failure
    node ← NODE(STATE=problem.INITIAL)
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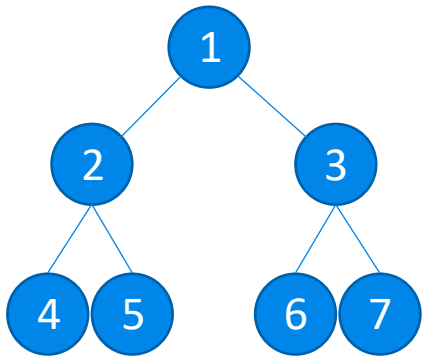
                add child to frontier
```

The problem:




Frontier: [2, 3]

Reached: [1, 2, 3]

Then 3, then onwards…

# Let's think about a general search architecture

```
function BEST-FIRST-SEARCH(problem, f) returns a solution node or failure
    node ← NODE(STATE=problem.INITIAL)
    frontier ← a priority queue ordered by f, with node as an element
    reached ← a lookup table, with one entry with key problem.INITIAL and value node

    while not IS-EMPTY(frontier) do
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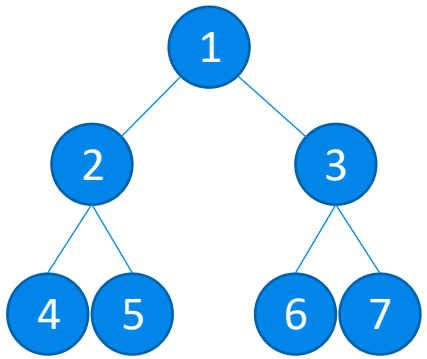
        for each child in EXPAND(problem, node) do
            s ← child.STATE
            if s is not in reached or child.PATH-COST < reached[s].PATH-COST then
                reached[s] ← child
                add child to frontier
    return failure
```

The problem:




Frontier: []

Reached: [...]

# What does it mean to expand?

```
function EXPAND(problem, node) yields nodes
    s ← node.STATE
    for each action in problem.ACTIONS(s) do
        s' ← problem.RESULT(s, action)
        cost ← node.PATH-COST + problem.ACTION-COST(s, action, s')
        yield NODE(STATE=s', PARENT=node, ACTION=action, PATH-COST=cost)
```

# What does it mean to expand?

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function EXPAND(problem, node) yields nodes
    s ← node.STATE
    for each action in problem.ACTIONS(s) do
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```

Your current state

# What does it mean to expand?

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```

Your current state

All the things you could do in the state

# What does it mean to expand?

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function EXPAND(problem, node) yields nodes
    s ← node.STATE
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        yield NODE(STATE=s', PARENT=node, ACTION=action, PATH-COST=cost)
```

Your current state

All the things you could do in the state

Where you could end up

# What does it mean to expand?

**function** EXPAND(*problem, node*) **yields** nodes

s ← *node*.STATE

**for each** *action* **in** *problem*.ACTIONS(*s*) **do**

*s'* ← *problem*.RESULT(*s, action*)

*cost* ← *node*.PATH-COST + *problem*.ACTION-COST(*s, action, s'*)

**yield** NODE(STATE=s', PARENT=*node*, ACTION=*action*, PATH-COST=*cost*)

Your current state

All the things you could do in the state

Where you could end up

How expensive it would be to get there

# What does it mean to expand?

```
function EXPAND(problem, node) yields nodes
    s ← node.STATE
    for each action in problem.ACTIONS(s) do
        s' ← problem.RESULT(s, action)
        cost ← node.PATH-COST + problem.ACTION-COST(s, action, s')
        yield NODE(STATE=s', PARENT=node, ACTION=action, PATH-COST=cost)
```

Your current state

All the things you could do in the state

Where you could end up

How expensive it would be to get there

generates a sequence of values, each time yield is encountered

# Uninformed Search

# A quick review of BFS, Dijkstra, and DFS

- We're going to go quickly through EECS 281 topics
- For a more detailed review of them, please see the slides of this week's discussion section
- Helpful document to compare all search algorithms we'll look at:
  - Canvas > Files > Lecture Slides > Search Algorithms at a Glance.pdf

# Uninformed Search

- Information available to algorithm
  - Only what is provided by problem definition
  - No information about how close a state is to the goal state

Note: BFS = Breadth-first search

*check that the root isn't a goal

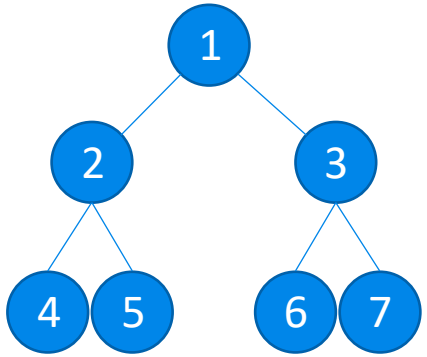
# BFS in the general search architecture

```
function BREADTH-FIRST-SEARCH(problem, f) returns a solution node or failure
    node ← NODE(STATE=problem.INITIAL)
    frontier ← a first-in-first-out queue, with node as an element*
    reached ← a set of states, with one entry with value node.STATE

    if problem.IS-GOAL(node.STATE) then return node
    while not IS-EMPTY(frontier) do
        node ← POP(frontier)
        for each child in EXPAND(problem, node) do
            if problem.IS-GOAL(child.STATE) then return child
            s ← child.STATE
            if s is not in reached or child.PATH-COST < reached[s].PATH-COST then
                reached[s] ← child
                add child to frontier

    return failure
```

The problem:

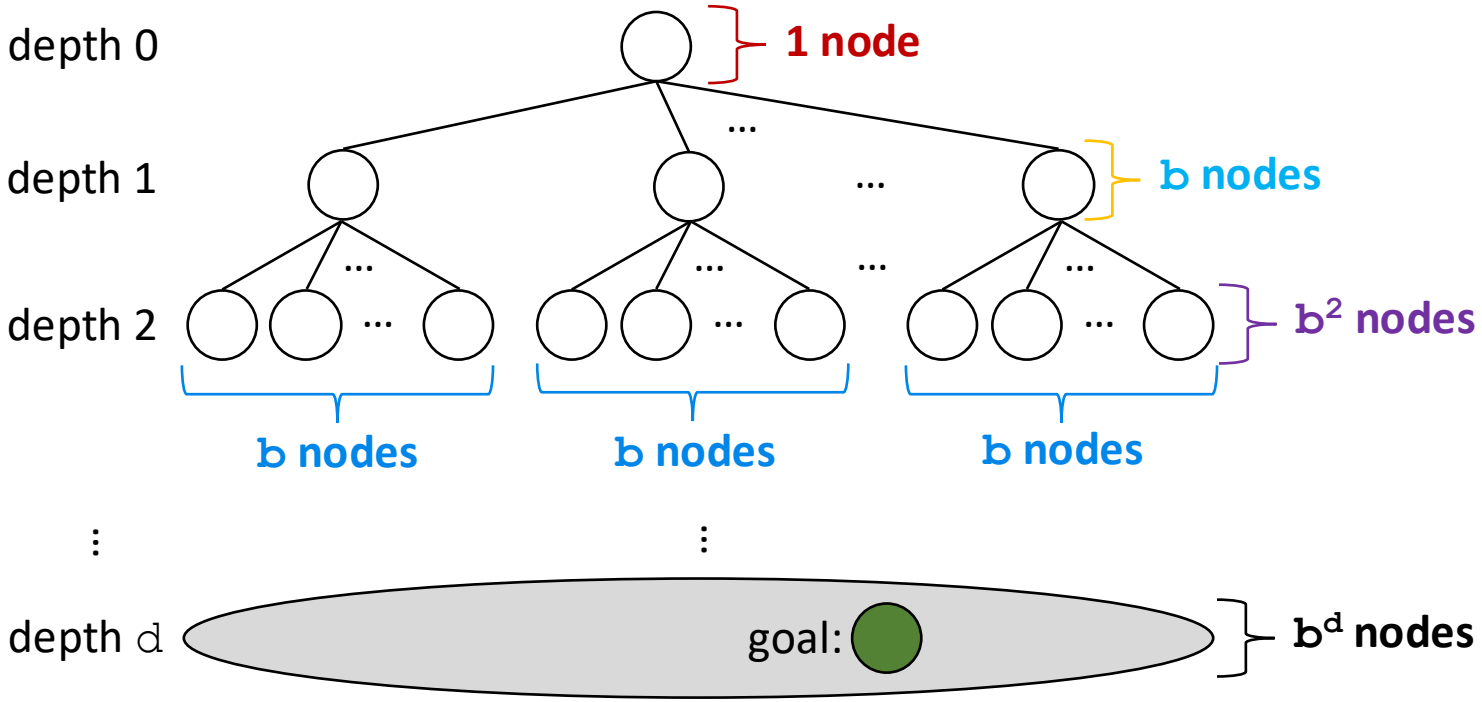



Early **goal test** – check if a node is a solution as soon as it is *generated*, rather than waiting until it is popped off the queue.

**Reached:** stores states
**Frontier:** first-in-first-out queue

# BFS: Time Complexity

- Assumptions:
  - We have a uniform branching factor, b
  - The depth of the shallowest solution is d




- BFS generates complete search trees at depth ≤ d

- What if BFS checked for goal state when expanding nodes?

Worst case number of nodes generated:

$1 + b + b^2 + \ldots + b^d = O(b^d)$

Thanks to ***early goal test***, the moment we generate the goal , we return.
If late goal test, then space is $O(b^{d+1})$ because d+1 nodes will be entered to frontier

# BFS Space Complexity

- Follows the analysis in the previous slide
- BFS stores all nodes (frontier)
  - Frontier → $O(b^d)$
- Space complexity is $O(b^d)$ thanks to early goal testing
- Generally, BFS runs out of space before it runs out of time!

# BFS: Search Performance Criteria

- Is BFS optimal?
  - BFS always find solution with minimal number of actions, because when it is generating nodes at depth *d*, it has already generated all the nodes at depth *d-1*
  - If all the actions have the same cost, it is optimal!
  - If all the actions don't have the same cost, it may not be optimal

- Is BFS complete?
  - Yes – it systematically searches through all possible paths at each depth

# BFS Limitations

- Time complexity is a problem
- Space complexity is the killer!

| Depth | Nodes | Time | Memory |
|---|---|---|---|
| 2 | 110 | 0.11 milliseconds | 107 kilobytes |
| 4 | 11,110 | 11 milliseconds | 10.6 megabytes |
| 6 | $10^6$ | 1.1 seconds | 1 gigabyte |
| 8 | $10^8$ | 2 minutes | 103 gigabytes |
| 10 | $10^{10}$ | 3 hours | 10 terabytes |
| 12 | $10^{12}$ | 13 days | 1 petabyte |
| 14 | $10^{14}$ | 3.5 years | 99 petabytes |
| 16 | $10^{16}$ | 350 years | 10 exabytes |

- Rule of thumb: exponential-complexity search problems **cannot** be solved by uninformed methods for any but the smallest instances!

# Uniform-Cost Search (Dijkstra's algorithm)

- Remember, BFS is optimal if all actions have the same cost

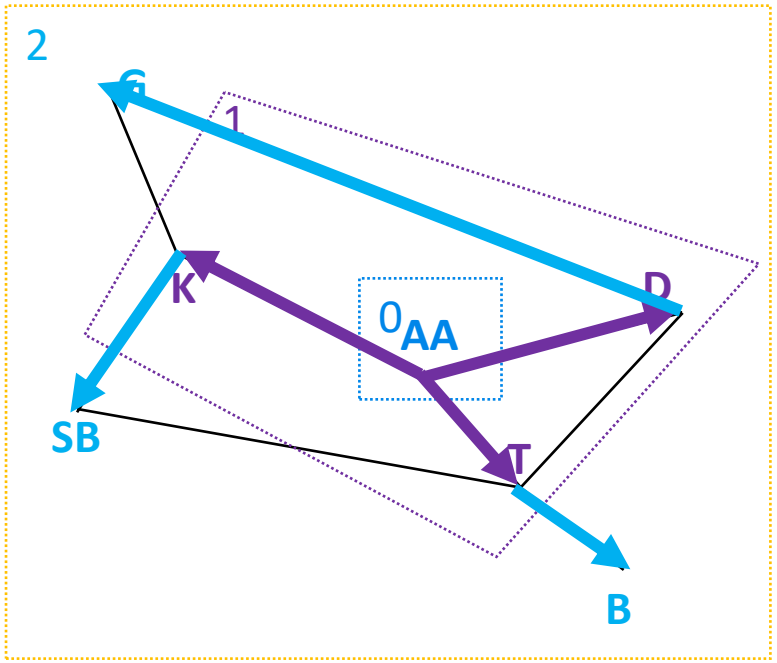
- Let's extend the algorithm so that it is optimal even if all actions do not have the same cost!

- Uniform-Cost Search
  - Expands node *n* with the lowest path cost *g(n)*
  - Frontier is stored in a priority queue ordered by *g*

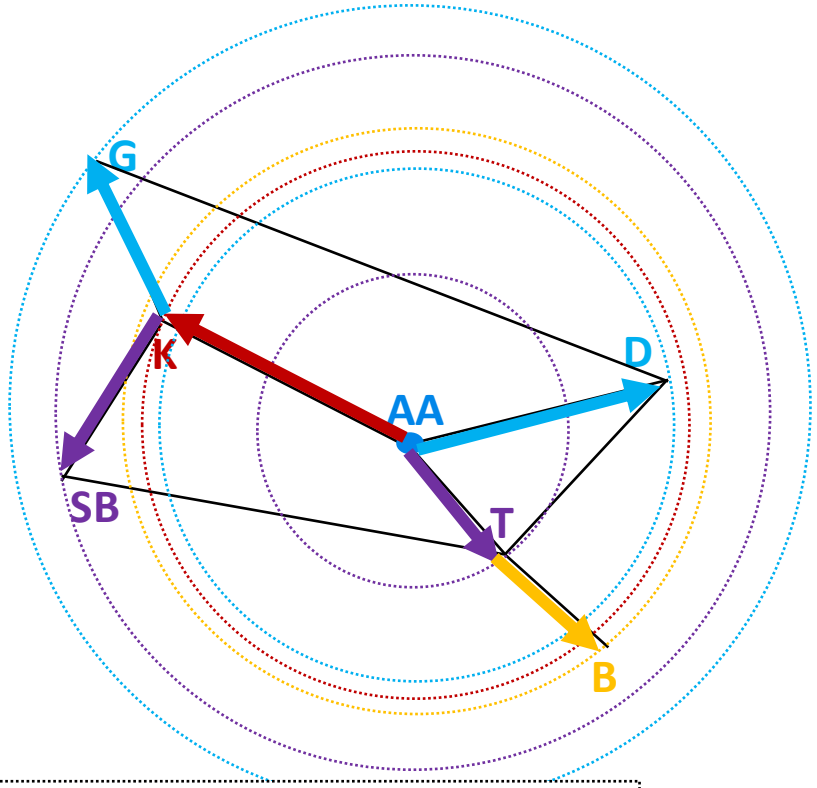
# BFS vs. UCS

Let's consider an image that is scaled by path cost.

BFS
Expand nodes in FIFO manner




UCS
Expand node on frontier with lowest path cost




UCS uses a priority queue!

# UCS in the general search architecture

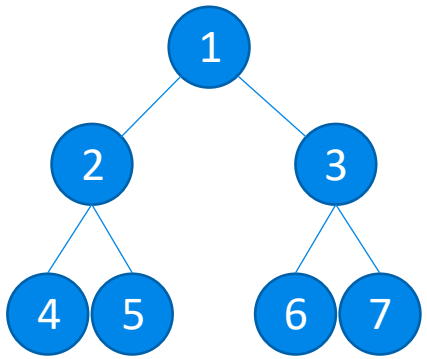
```
function UNIFORM-COST-SEARCH(problem, f) returns a solution node or failure
    node ← NODE(STATE=problem.INITIAL)
    frontier ← a priority queue ordered by f, with node as an element
    reached ← a lookup table, with one entry with key problem.INITIAL and value node

    while not IS-EMPTY(frontier) do
        node ← POP(frontier)
        if problem.IS-GOAL(node.STATE) then return node
        for each child in EXPAND(problem, node) do
            s ← child.STATE
            if s is not in reached or child.PATH-COST < reached[s].PATH-COST then
                reached[s] ← child
                add child to frontier

    return failure
```

**Late goal test** – checking whether a node is a solution node when it is popped off the queue

The problem:

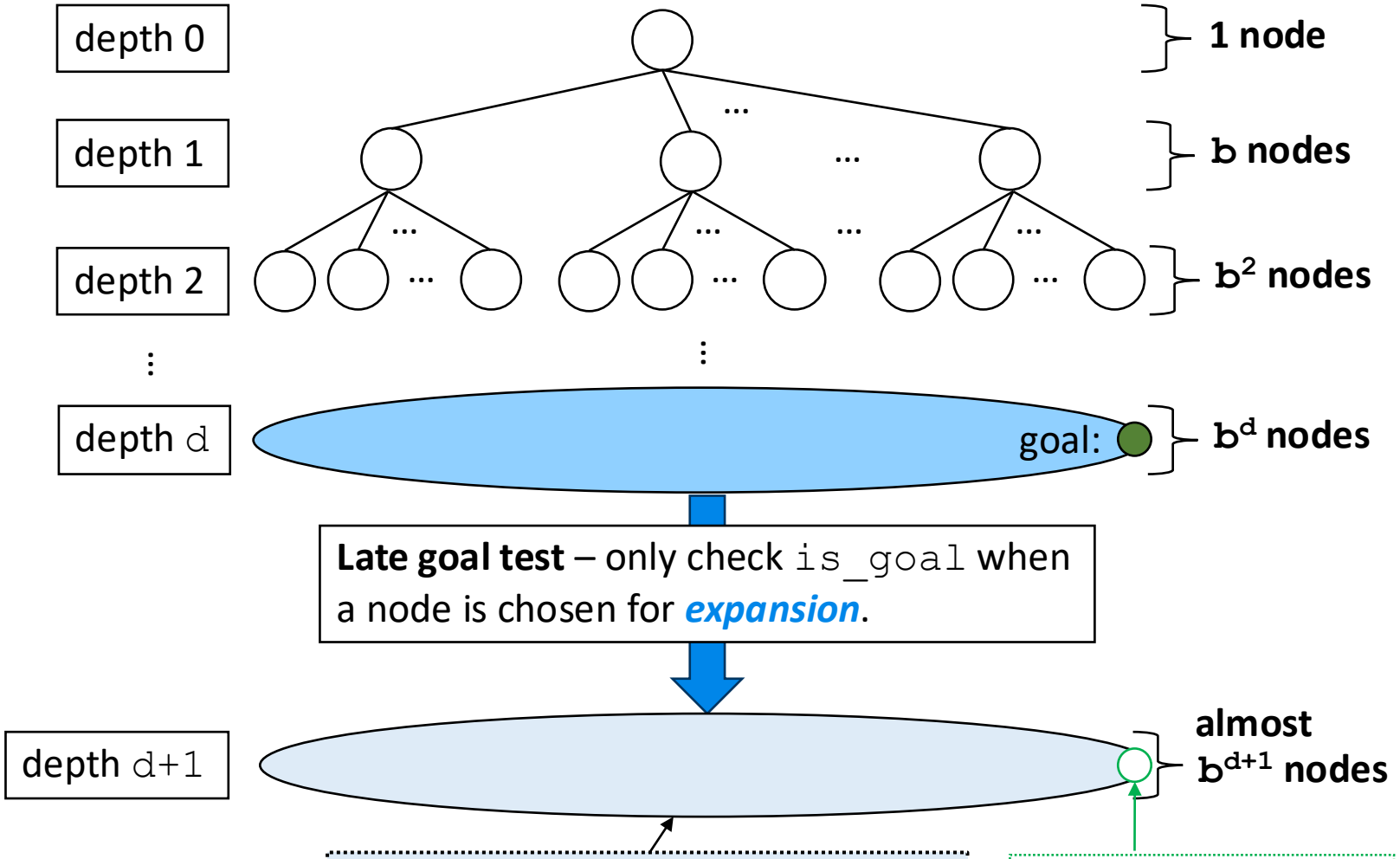



**Reached:** stores nodes
**Frontier:** priority queue, path-cost

## Time and Space Complexity

- Worst case, each step has minimum cost $\epsilon$.

C* = the cost of the optimal solution
$\epsilon$ = the minimum step-cost




So: $d = C^*/\epsilon$
in the worst case

Or if $\epsilon \nmid C^*$ then $d = \lfloor C^*/\epsilon \rfloor$

**Late goal test** – only check `is_goal` when a node is chosen for ***expansion***.

sucessors of non-goals at depth `d` that were reached

goal's sucessors not reached.
Let `gs` = number of these.

Worst case number of nodes generated:

$$\left(\sum_{i=0}^{d+1} b^i\right) - gs = O\left(b^{d+1}\right) = O\left(b^{\lfloor C^*/\epsilon \rfloor + 1}\right)$$

# UCS: Search Performance Criteria

- Is UCS Optimal?
  - Yes! When node *n* is expanded, the optimal path to *n* has been found
  - (Assuming step-costs are non-negative)

- Is UCS Complete?
  - Yes, assuming all step costs are greater than a small constant, epsilon $\varepsilon$

- Time and Space Complexity
  - $C^*$ = the cost of the optimal solution
  - $\varepsilon$ = the minimum step-cost
  - $O(b^{1+[C^*/\varepsilon]})$

# DFS in the general search architecture

Note: We assume there are no cycles and do not save a *reached* set (saves memory!)

```
function DEPTH-FIRST-SEARCH(problem, f) returns a solution node or failure
    node ← NODE(STATE=problem.INITIAL)
    frontier ← a last-in first-out queue, with node as an element

    while not IS-EMPTY(frontier) do
        node ← POP(frontier)
        if problem.IS-GOAL(node.STATE) then return node
        for each child in EXPAND(problem, node) do
            s ← child.STATE
            add child to frontier
    return failure
```

**Late goal test** – checking whether a node is a solution node when it is popped off the queue

**Reached:** not necessary
**Frontier:** last-in-first-out queue

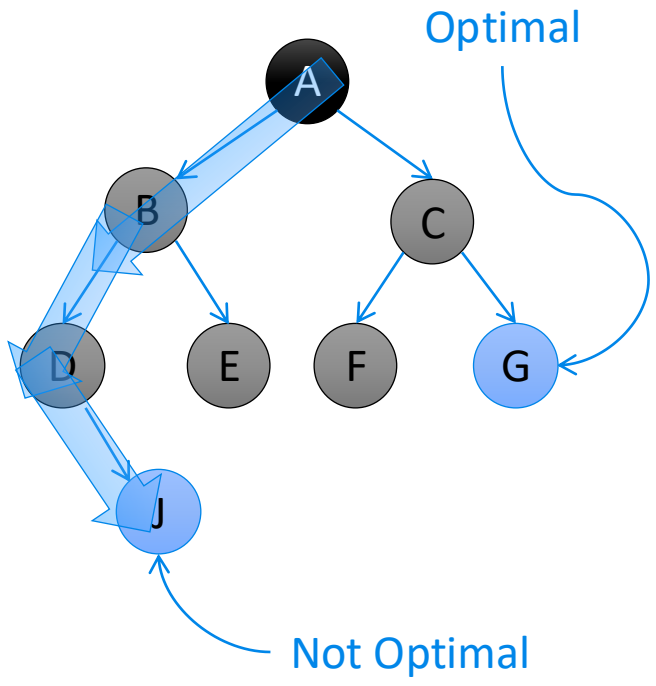
# DFS saves memory

- Why?
  - No reached structure
  - DFS often applied to problems where cycles are not present or rare
- What if there are cycles?
  - Follow chain of parent pointers up the tree to see if state at the end of the path has appeared earlier

# DFS: Search Performance Criteria

- Is DFS Optimal?
  - No - returns the first solution it finds, even if it's not the cheapest

- Is DFS Complete? It depends…
  - Complete: Graph search (finite spaces)
  - NOT Complete: Tree search (finite spaces)
  - NOT Complete: Both versions (infinite spaces)



# DFS Space Complexity

Let m = maximum depth of the **tree**. Maybe much bigger than d! (depth of sol)

depth 0 — 1 node

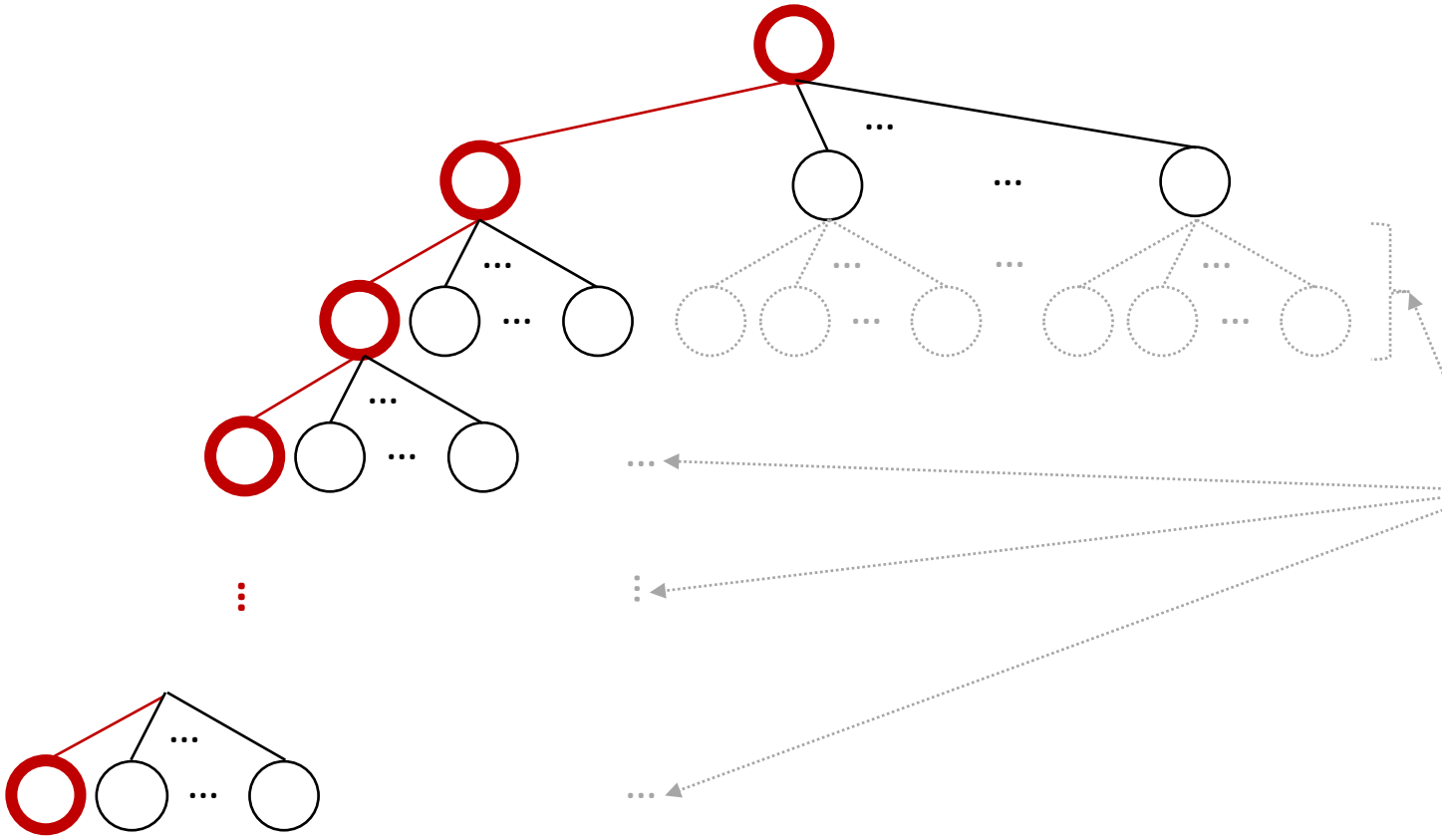
depth 1 — b nodes

depth 2 — b nodes

depth 3 — b nodes

⋮

depth m — b nodes



Nodes that won't be reached until later (not generated yet).

Worst case, number of nodes in memory at one time:

$$1 + \underbrace{b + b + \ldots + b}_{m \text{ of these}} = O(bm)$$

No maximum depth m? This is the case in infinite spaces
Then DFS is not complete and could use unbounded space.

# DFS Performance

**Space complexity**

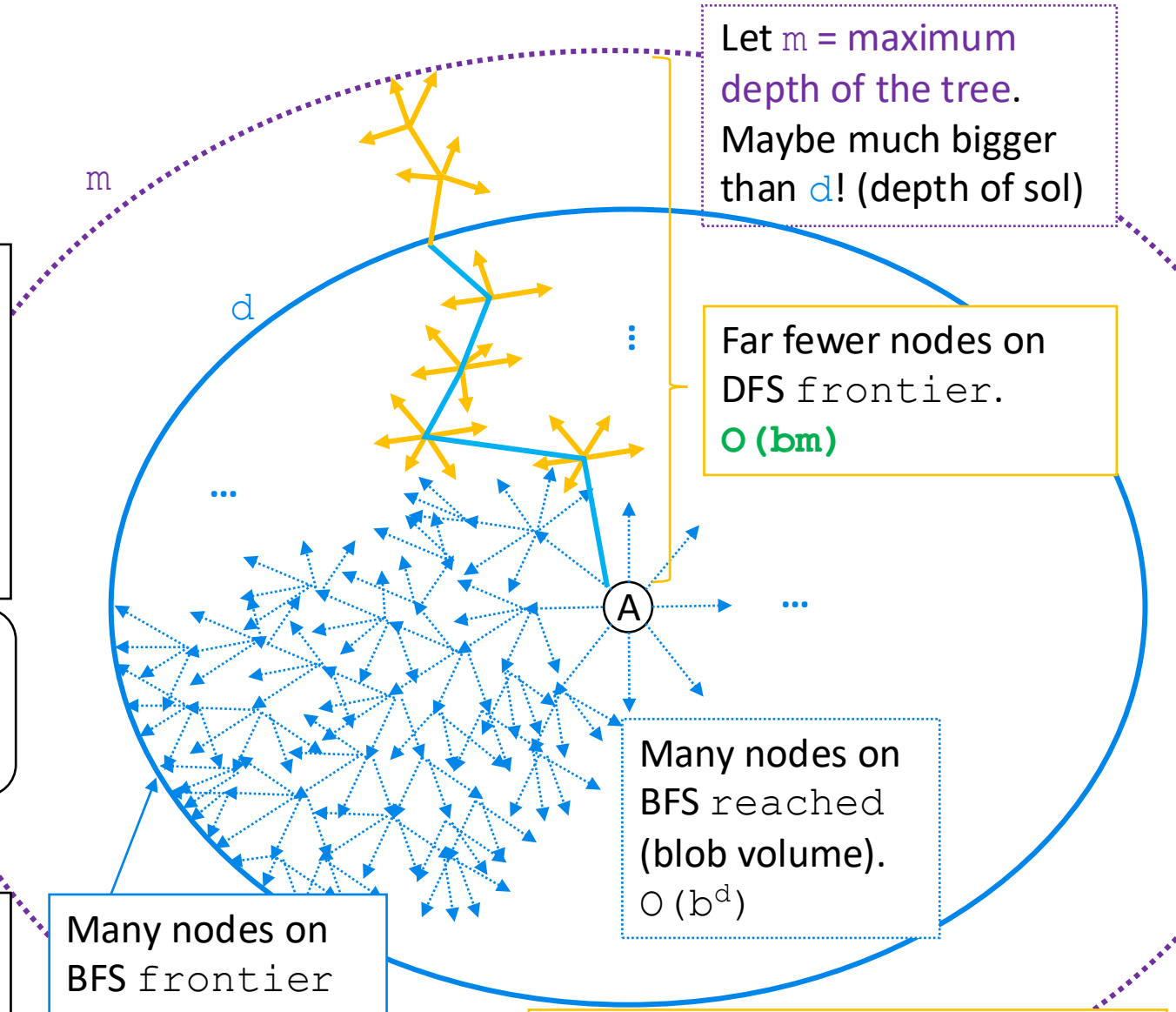
- When **tree-like search** is feasible, DFS requires ***far less memory*** than BFS or UCS:
  - No `reached`
  - `frontier` is much smaller

States in mem. for `b` = 10, `d` = 16, `m` = 100:

- BFS: $O(b^d)$ 10,000,000,000,000,000
- DFS: $O(bm)$ 1,000 ← **Huge Savings!**

**Time complexity**

- $O(b^m)$ – still may have to search the entire space to the maximum depth in the tree. (Still have to "fill up the blob", just not *saving* those nodes.)




Let `m` = maximum depth of the tree. Maybe much bigger than `d`! (depth of sol)

Far fewer nodes on DFS `frontier`. $O(bm)$

Many nodes on BFS `reached` (blob volume). $O(b^d)$

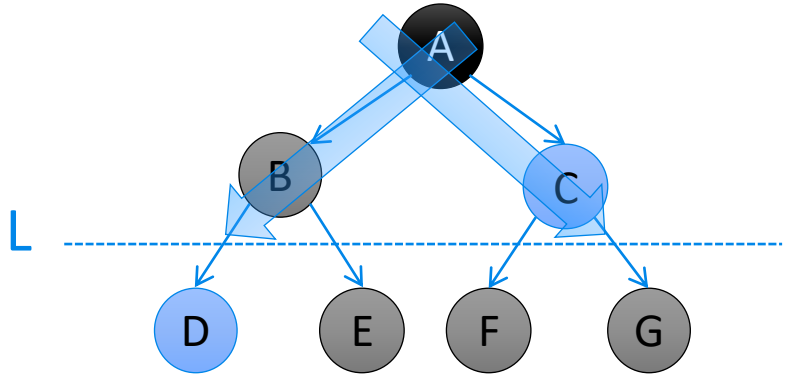
Many nodes on BFS `frontier` (blob perimeter). $O(b^d)$

Due to low memory usage, DFS tree-like search is at the heart of many AI techniques, like constraint satisfaction and logic.

# How Can We Make DFS Usable?

- Problem: Unbounded trees!

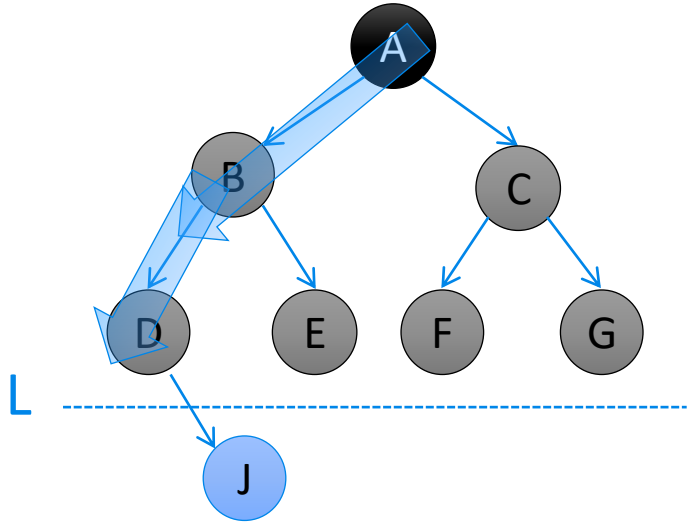
- Solution – Depth-Limited Search:
  - Identify a predetermined depth limit, *L*
  - Run DFS and ignore any successors beyond *L*



# Limitations of DFS-Limited

- An incorrect choice of L dooms us!

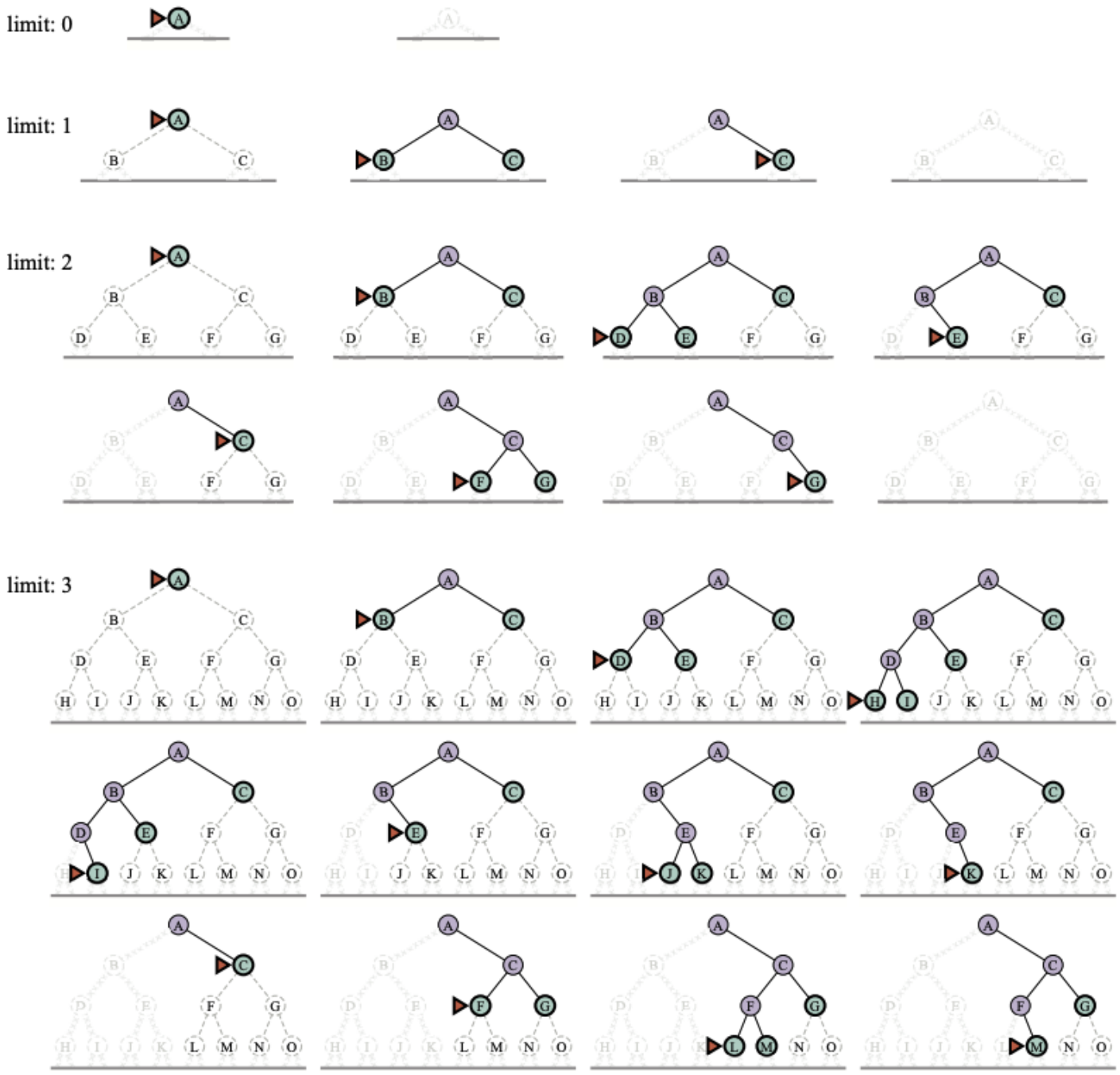
- Question: how should we choose L?
  - Well, what do we want it to do?



# Iterative Deepening Search (IDS)

- Our DFS problem:
  - We won't find a solution if d > L
- Our DFS solution:
  - Iteratively increase L
  - Stop when the shallowest goal node has been reached
- Analysis:
  - Complete if branching factor is finite
  - Optimal for problems where all actions have the same cost
  - Space complexity: O(bd) – same as DFS for depth d!
  - Question: how wasteful is it to regenerate trees?

# IDS



# IDS: Time Complexity

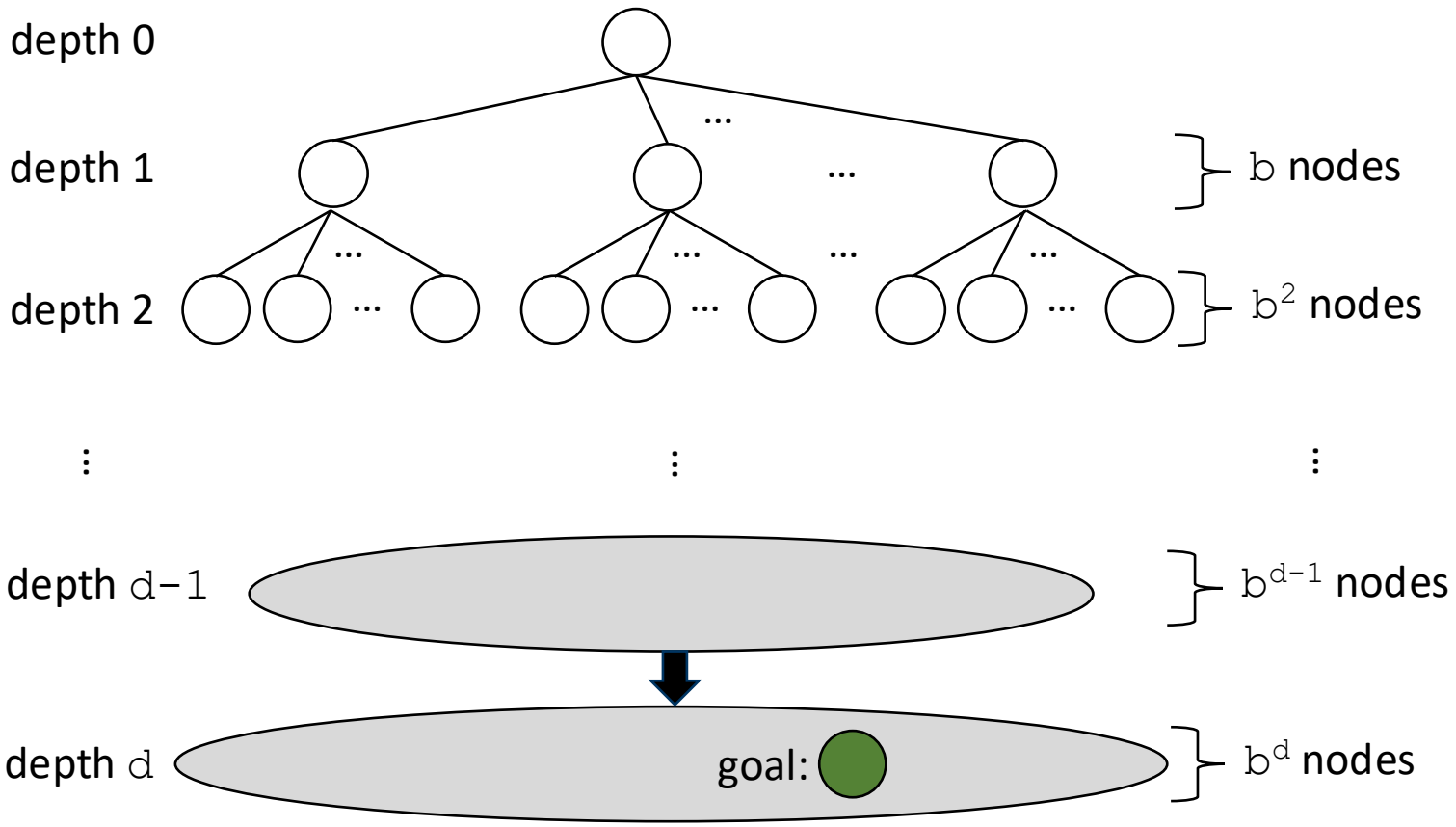
- Proportionately, most nodes are at the bottom of the tree

- Node regeneration
  - Most nodes are at the bottom of the tree
  - Nodes at bottom are generated once → d
  - Nodes at next lowest level are generated twice → 2(d-1)
  - Nodes at the top are generated d times → d

- Total number of generated nodes:
  $N(IDS) = (d) * b + (d-1) * b^2 + \ldots + (1)\, b^d$

- Time Complexity: $O(b^d)$ ← Asymptotically the same as BFS!

# How many times does IDS generate each level?




| times generated |
|---|
| d |
| d-1 |
| ⋮ |
| 2 |
| 1 |

**Space complexity**:
Same as DFS: **O (bd)**
If we stored nodes to save regeneration, space complexity would be like BFS!

Number of nodes generated (**time complexity**):
$db + (d-1)b^2 + (d-2)b^3 + \ldots + 2b^{d-1} + b^d = \mathbf{O(b^d)}$

Asymptotically the same as BFS!

# Summary of Uniformed Search Methods

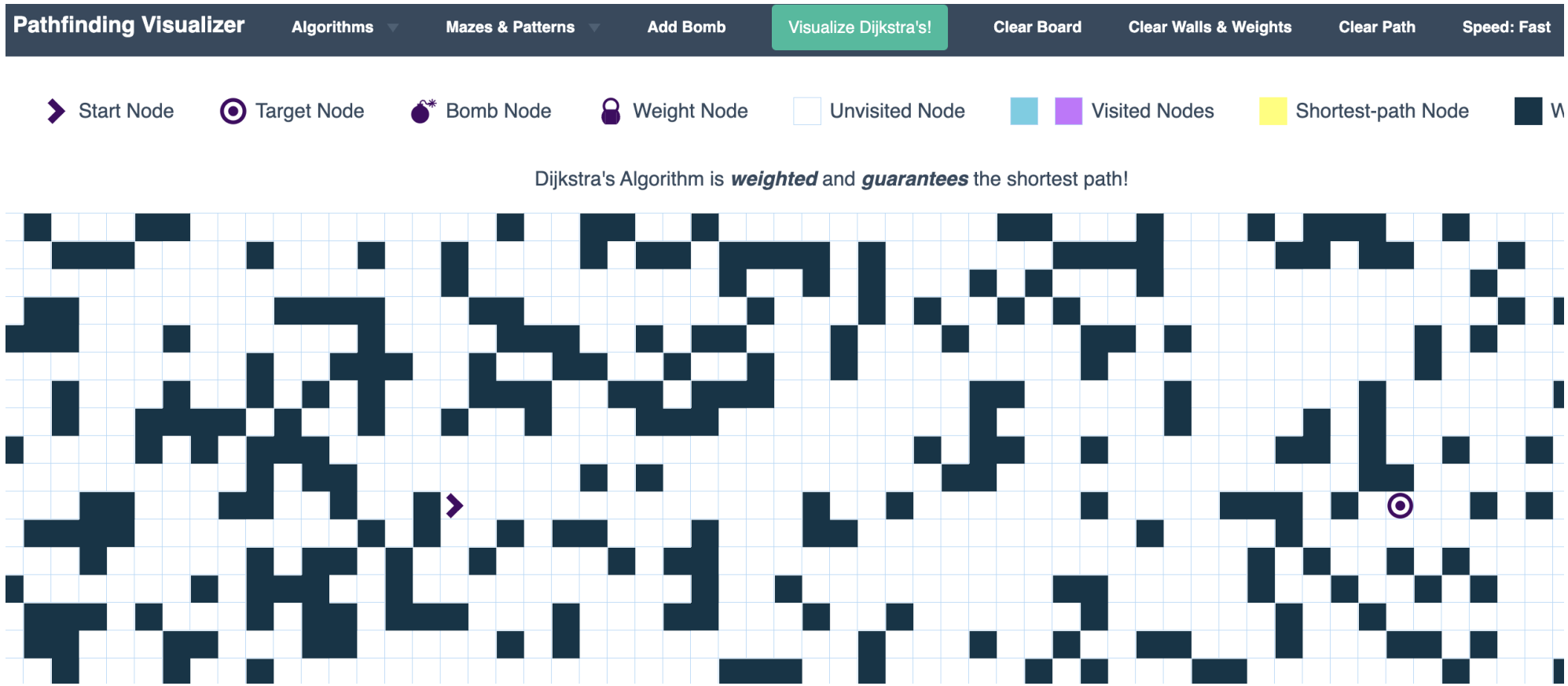
| Criterion | Breadth-First | Uniform-Cost | Depth-First | Depth-Limited | Iterative Deepening |
|---|---|---|---|---|---|
| Complete? | Yes[1] | Yes[1,2] | No | No | Yes[1] |
| Optimal? | Yes[3] | Yes | No | No | Yes[3] |
| Time Complexity | $O(b^d)$ | $O(b^{1+(C^*/\varepsilon)})$ | $O(b^m)$ | $O(b^l)$ | $O(b^d)$ |
| Space Complexity | $O(b^d)$ | $O(b^{1+(C^*/\varepsilon)})$ | $O(bm)$ | $O(bl)$ | $O(bd)$ |

Exponential in Depth

Linear in Depth

1. Complete if b is finite and the state space either has a solution or is finite
2. Complete if all action costs are $\geq \epsilon > 0$
3. Cost-optimal if action costs are identical

These caveats are important!

# Demo!

https://clementmihailescu.github.io/Pathfinding-Visualizer/